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Intelligent Traffic Signal Control Using Fuzzy Logic, Multi-Agent Systems, and Optimization Techniques: A Comprehensive Study

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Abstract—The problems caused by growing numbers of vehicles and rapid urban development in first-tier Indian cities are now commonplace. Managing traffic lights in the traditional way is inefficient and continues to worsen the problem by increasing congestion and wait times, wasting fuel, and polluting the environment. Integrated traffic control systems can utilize IoT and Intelligent Transport Systems (ITS) to overcome these challenges. This paper focuses on the design and performance analysis of an easily scalable and cost effective IoT sensor architecture for smart traffic light control within the budget and infrastructural constraints of Indian metropolitan cities. The system utilizes economical sensors, edge controllers, efficient embedded systems, value engineering in communications, and adaptive signal processing for traffic management based on the actual density of vehicles. For modularity and positive engineering in the extension and the utilization of artificial intelligence optimization techniques, the system employs a scalable, fault-tolerant modular IoT stack. The results show an average reduction in intersection throughput and emission proxies, as well as average waiting times and queue lengths. This confirms the proposed method, evaluated against SUMO microscopic traffic simulations, is effective compared to the conventional fixed-time signal method. The results support the adaptive, IoT-driven traffic control approach as a viable and effective smart city solution for emerging urban areas.

Keywords—Smart Traffic Lights; Internet of Things (IoT), Intelligent Transportation Systems, Cities, Urban Traffic Management

I. INTRODUCTION

One of the most urgent problems of modern cities is the urban traffic jams which cause loss of revenue, increase in pollution and a decrease in the quality of life. Although city planners employ a number of traffic signal control methods such as fixed-time and actuated control, they prove to be ineffective due to a lack of ability to address and respond to changes in real-time due to a traffic problem. In an effort to solve these problems, a number of traffic signal control systems which employ artificial intelligence to replicate human thinking and adaptive decision making have been proposed.

Due to randomness in the traffic density, the current traffic control system in the cities is inefficient. In a recent study of Delhi, it was reported that 10% of traffic fatalities in Delhi happen by vehicles running red lights. Up to 6,420 individuals were convicted for running red lights last year. Apart from this, delay in traffic flow causes fuel loss also. Everyday Delhi wastes 0.37 million kilograms of CNG, 0.13 million liters of diesel and 0.41 million liters of gasoline as an outcome of automobiles sitting idle. Engine exhaust, by petrol, diesel and gas, contains more than 40 hazardous air pollutants. Around 70% of the air pollution comes directly from the emissions from vehicles. Idling cars and slow speeds result in four to eight times greater emissions each journey. Currently, a predetermined timing method is used to control signals. It causes traffic jams during rush hour and encourages cars to avoid signals at slow time. If the timer of traffic signal can be programmed according to the issue of traffic congestion can be drastically minimized due to the constantly fluctuating traffic density.

Currently, signals are managed through a preset timing system. It results in pileups during the busiest time and also induces drivers to skip signals during lean hours. If the timer of traffic signal can be programmed according to the issue of traffic congestion can be greatly lessened due to the constantly fluctuating traffic density and average increase of quantity of vehicles on road by 20 percent per year, modern societies even with well-planned road management system and enough infrastructure, still face traffic congestion problem. Traffic congestion gives extra time consumption during travel which results in huge societal and economical losses. Traffic problem is mainly due to the three reasons; ever-increasing number of vehicles on road, lack of adequate road infrastructure and a conventional traffic light system. First two reasons have no solution due to economic, environmental and to some extent political concerns. In this work we focus on the third approach, i.e. conventional traffic signal system with a special focus on freeway traffic management and control.

This paper proposes a smart traffic controller based on fuzzy logic using MATLAB 8.1 and application of sensor-based Internet of Things (IoT) is presented so as to real time adapt the four (may be lesser too, say two to three) inputs and to control the green time duration at the traffic junction in order to reduce the traffic jams and hence to save the time and money of motor vehicle.

II. LITERATURE REVIEW

Fuzzy logic traffic controllers use qualitative variables like queue length, time, and traffic density to manage signal phases. Niittymäki [1] laid the first milestone for fuzzy traffic signal control, while similarity logic reasoning was later proposed to improve the precision of reasoning under uncertainty [2]. Genetic fuzzy logic controllers (GFLC) further enhanced adaptability by iteratively optimizing fuzzy rule sets [3].

Advanced multi-phased fuzzy controllers reduced delay and queue length at urban expressways and isolated intersections successfully. [4]. Extending these concepts to traffic networks, anti-congestion fuzzy algorithms mitigated congestion at several intersections [5]. The applicability and the robustness of the fuzzy techniques in various traffic environments were noted in several reviews and surveys [6].

Type-2 fuzzy logic controllers were designed to cope with increased uncertainty in traffic systems, and compared to type-1 fuzzy systems, they improved the management of sensor noise and variable traffic flow [7]. Due to the integration of expert knowledge and historical data, the knowledge-based models gained in interpretability and decision-making reliability [8].

Large interconnected urban systems present challenges for centralized control systems. In multi-agent systems (MAS), each autonomous agent makes decisions for a specific intersection [9]. For example, distributed geometric fuzzy multi-agent models improved coordination and reduced congestion on arterial networks [10]. These strategies fit the smart city model, emphasizing systems that allow autonomy and collaboration.

To reduce vehicle idle time, fuel usage, and emission, several optimization techniques have been employed, including particle swarm optimization (PSO) and evolutionary algorithms [11]. Systems that use swarm intelligence for real-time signal control adjust signals in response to changing conditions and show greater control performance than static signal control [12]. In terms of smoothing and responsiveness, hybrid systems that integrate fuzzy logic and optimization techniques also demonstrate good results [13].

III. Research Methodology

The generalized methodology for intelligent traffic signal control featured in the reviewed literature can be broken down into the following steps:

Data Collection: Vehicle data (total amount of vehicles, length of queues, speed, and time spent waiting) is processed using Internet of things (IoT) sensors, cameras, and embedded detectors.

Data Filtering and Estimation of Traffic State: The raw data is worked on through filtering, and then the data is processed into linguistic variables, or other numerical values that intelligent controllers can work.

Implementation of a Control Action Strategy: Fuzzy logic controllers signal changes in the length of the time a signal is green or red according to the traffic flow in real time. Multi-agent systems provide coordination on an inter-traffic signal level for network level optimization.

Control parameters are again processed using the optimization equations to achieve the closest values to the ideal: minimization of average delay, fuel consumption, and emission of pollutants.

The system is evaluated based on the following criteria: average delay, queue length, throughput, fuel consumption, and emission levels in the case of evolving control patterns, self-organizing or other traffic patterns the control auto.

IV. Mathematical Modeling

A. Traffic State Representation

Let

$Q_i(t)$ = queue length on approach i at time t

$D_i(t)$ = average vehicle delay

$\lambda_i(t)$ = arrival rate

$\mu_i(t)$ = service rate

Queue dynamics are modeled as:

$$Qi(t+1) = Qi(t) + \lambda i(t) - \mu i(t) \tag{1}$$

B. Fuzzy Logic Control Model

Fuzzy inputs:

Queue length (Low, Medium, High)

Waiting time (Short, Moderate, Long)

Fuzzy output:

Green signal extension time

Rule example:

IF Qi is High AND Di is Long THEN Gi is Maximum

C. Optimization Objective Function

The signal timing optimization problem is formulated as:

$$Min_J = \sum_{i=1}^{N} \quad W_1 1 D_i + W_2 Q_i + W_3 E_i \tag{2}$$

where

Ei = fuel consumption/emission component

w1, w2, w3 are weighting coefficients by the Swarm intelligence or evolutionary algorithms iteratively update signal timings to minimize J

TABLE I. INPUT VARIABLES AND FUZZY SETS OF FUZZY-BASED ALGORITHM

Input Variables	Fuzzy Sets
Queue_length	{very_less, less, moderate, long}
Arrival_rate	{very_less, less, moderate, large, very large}
Rem_time_green	{very_less, less, moderate, large, very large}
Peak_hours	{very_light, heavy_morning, moderate, heavy_evening, light_evening}
Time_extension	{more_decrease, decrease, no_change, increase, more_increase}

V. Theory And Mathematical Modeling Of Intelligent Traffic Signal Control

This section explains the theory and mathematics of intelligent traffic signal control problems within the context of optimization, fuzzy inference, and traffic flow modeling.

A. Traffic Flow and Queue Dynamics

An urban road network can be defined as N signalized intersections. For each intersection i, characteristics of the traffic situation are defined using macroscopic traffic flow.

Let:

- Qi(t) be Queue length (number of vehicles) on approach i at time t
- λi(t) be Vehicle arrival rate (number of vehicles per second)

- $\mu i(t)$ be the Service rate controlled by the duration of the green signal
- $gi(t)$ be effective green time

For the queue, the following evolution expression is used:

$$Qi(t+1) = max[0, Qi(t) + \lambda i(t)\Delta t - \mu i(t)gi(t)] \tag{3}$$

This expression captures the congestion accumulation and dissipation over successive signal cycles.

B. Delay and Performance Modeling

Average vehicle delay Di at intersection i is computed as:

$$D_i = \frac{1}{Q_i}\sum_1^{Q_i} \quad t_{Departure}{}^k - t_{DArrival}{}^k \tag{4}$$

Total network delay:

$$D_{total} = \sum_{i=1}^{n} \quad D_i \tag{5}$$

These metrics are widely used to evaluate traffic signal control efficiency in simulation-based studies.

C. Fuzzy Logic Control Model

Fuzzy logic controllers map traffic states into control actions using linguistic reasoning.

Input Variables:

- Queue Length: {Low, Medium, High}
- Waiting Time: {Short, Moderate, Long}

Output Variable:

- Green Time Adjustment: {Decrease, Maintain, Extend}

A representative fuzzy rule:

IF Queue Length is High AND Waiting Time is Long THEN Green Time is ExtendedFuzzy inference systems have four components: fuzzification, rule evaluation, aggregation, and defuzzification.

D. Optimization Objective Function

In order to enhance the optimization of signal timing for the overall performance of global traffic, the problem has been framed as a multi-objective function.

For each signal i in I {I = 1, 2, …, N}, we have the following:

$$Min_J = \sum_{i=1}^{N} \quad W_1 1D_i + W_2Q_i + W_3E_i \tag{5}$$

Where Ei corresponds to the emissions or fuel component of the environment w1 , w2 , w3 are the balanced weighting variables The parameters of the traffic signals are optimized using swarm intelligence or evolutionary type of algorithms for the purpose of reducing J

VI. SUMO-Based Simulation Setup

Simulation Environment We employ the Simulation of Urban Mobility (SUMO), one of the most popular microscopic traffic simulators, to evaluate the proposed intelligent traffic signal control system. Simulation Parameters:

Parameter	Value
Simulator	SUMO 1.x
Simulation Time	3600 s
Time Step	1 s
Traffic Dema	Variable (Low–High)
Intersection Type	Four-leg isolated & networked
Vehicle Types	Passenger cars
Control Strategies	Fixed-time, Actuated, Fuzzy,Hybrid

VII. Simulation Results And Analysis

A. Average Waiting Time

The average waiting time comparison of the different algorithms is described in table 1.

TABLE II. Describe the Average Waiting Time

Fixed-Time	68.4
Actuated	54.7
Fuzzy Logic	39.2
Hybrid Intelligent	31.6

B. Average Queue Length

The Average queue length of the vehicles is described in the table 2.

TABLE III. Describe the Average Queue Length

Control Method	Queue Length (Veh)
Fixed Time	21.3
Actuated	17.8
Fuzzy Logic	12.4
Hybrid Intelligent	9.1

C. Throughput Performance

The throughput performance of the vehicles is described in table 3.

Table III: Describe the Throughput performance

Control Method	Throughput (veh/hr)
Fixed-Time	1850
Actuated	2010
Fuzzy Logic	2245
Hybrid Intelligent	2410

D. Fuel Consumption

Control Method	Fuel Consumption (ml/veh)
Fixed-Time	78.6
Actuated	65.2
Fuzzy Logic	52.9
Hybrid Intelligent	47.3

The proposed model demonstrates good performance, achieving a reduced average queue length, increased through put and optimized fuel consumption compared to the base line system.

VIII. Conclusion

Computer simulation was being carried out using MATLAB 8.1 and the fuzzy logic tool box. Fuzzy logic tool box served two purposes one usefulness for quickly building the required rules and any changes required later can be incorporated easily, secondly reduction in the development time of the simulation model. Simulation was carried out for two different cases; taking all four input parameters that produces a single output variable and taking any three input parameters that produces a single output variable. This has been done so as to calculate the exact weightage of each input parameter. In real time implementation all four input parameters may not be possibly deployed due to cost and complexity reasons. Comparison between these two cases gave a guideline about the weightage of each input parameter.

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